A Triple-Branch Architecture With Multiscale Attention for Spatiotemporal Remote Sensing Fusion

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***Abstract*—Existing remote sensing spatiotemporal fusion methods often suffer from excessive smoothing at object boundaries, insufficient modeling of global contextual information, and underutilization of temporal change features. To improve these issues, a triple-branch convolutional neural network is proposed, consisting of a dual-stream spatial network and a temporal difference network. Within the spatial branches, a spatiotemporal adaptive modulation (STAM) module is integrated, in which spatial attention is combined with squeeze-and-excitation (SE) channel attention to enhance feature discrimination. In the temporal branch, lightweight depthwise separable convolutions (DSCs) are employed to efficiently refine temporal difference features. Furthermore, a composite loss function is designed to simultaneously optimize pixel-level accuracy, structural fidelity, and edge detail restoration. Experiments on the CIA, LGC, and AHB datasets demonstrate that the proposed method achieves marked improvements over current models in reconstructing high-frequency edges and low-frequency structures, maintaining global semantic consistency, and suppressing dynamic change noise, with an average increase of 1.46 dB in PSNR and an average decrease of 0.15 in ERGAS.**

***Index Terms*—Spatiotemporal adaptive modulation (STAM), spatiotemporal fusion, squeeze-and-excitation (SE), triple-branch convolutional neural network.**

I. Introduction

CURRENT remote sensing satellite sensors are constrained by physical mechanisms and manufacturing costs, making it difficult to simultaneously acquire both high temporal and high spatial resolution Earth observation data. Instead, they produce either fine temporal and coarse spatial resolution (CR) images or coarse temporal and fine spatial resolution (FR) images. Specifically, multispectral sensors such as advanced very high resolution radiometer (AVHRR), moderate-resolution imaging spectroradiometer (MODIS), and sea-viewing wide field-of-view sensor (SeaWiFS) can achieve daily scale or even subdaily scale continuous observations, which are suitable for large-scale climate and vegetation index time-series analyses; however, their spatial resolution ranges only from 250 to 1000 m, leading to significant loss of spatial detail and an inability to distinguish small-scale land features. In contrast, sensors like Landsat, Sentinel-2, and advanced spaceborne thermal emission and reflection radiometer (ASTER) offer spatial resolutions of 10–30 m, facilitating fine-scale spatial feature interpretation, but their revisit intervals of 10 to 30 days—and susceptibility to cloud cover—create temporal blind spots for short-term change monitoring and rapid event response [1]. Nevertheless, high spatiotemporal resolution imagery is critically important in many application domains, including urban expansion monitoring [2], [3], [4], crop yield estimation [5], natural disaster assessment [6], [7], vegetation change detection [8], [9], and climate variability analysis [10]. Accordingly, in recent years, algorithms that fuse high temporal resolution coarse images with high spatial resolution fine images to produce data with both high temporal and spatial resolution have attracted widespread attention. Current spatiotemporal fusion methods can be categorized into unmixing-based, weight-function-based, Bayesian-based, learning-based, and hybrid methods [1].

Unmixing-based spatiotemporal fusion methods typically treat surface reflectance as a linear mixture of several pure endmembers. They first estimate the abundance of each endmember, then reconstruct the high-resolution image at the prediction date by applying the high-resolution endmember spectral library derived from a reference image, and finally perform residual correction. The multisensor multiresolution technique (MMT) was the first spectral unmixing-based fusion method [11], laying the theoretical and implementation foundations for subsequent unmixing models [12]. The linear spectral unmixing-based spatiotemporal data fusion model (LSUSDFM) linearly unmixed low-resolution pixels into endmember abundances, then reconstructed the high-resolution image using the spectral library from the reference date, and employed a sliding-window refinement of abundance estimates to mitigate boundary artifacts [13]. The object-based

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spatial unmixing model (OBSUM) performs object-based image analysis by segmenting homogeneous regions and conducting unmixing at the object level, followed by object-level and pixel-level residual corrections to generate predictions [14]. However, unmixing-based methods generally suffer from ill-posedness, are sensitive to noise and parameter choices, and often yield unstable abundance estimates [15].

The first spatiotemporal fusion approach employing a weight function is the spatial and temporal adaptive reflectance fusion model (STARFM) [16]. STARFM uses a reference-date Landsat image as a high-resolution spatial template and applies a weight function to distribute the reflectance change observed by MODIS between the reference and prediction dates onto high-resolution pixels, thereby estimating high-resolution reflectance on the prediction date. An enhanced STARFM (ESTARFM) overcomes the smoothing and error issues in heterogeneous landscapes of STARFM by introducing similar-pixel selection, conversion-coefficient correction, and temporal weighting, significantly improving high-resolution fusion accuracy [17]. GLM-STF uses the generalized linear model (GLM) to fit the pixel-value mapping between coarse- and fine-resolution data and constructs spatial and temporal weight functions to predict daily uncrewed aerial vehicle imagery [18]. However, weight-function-based methods perform poorly in heterogeneous regions: the pure-pixel assumption breaks down in areas with frequent spectral mixing or abrupt changes, leading to spectral distortion and spatial artifacts and limiting their ability to capture the trend of sudden changes.

Bayesian-based methods formalize spatiotemporal fusion as a Bayesian estimation task with uncertainty quantification. By specifying prior distributions, likelihood functions, and performing posterior inference, the fusion task becomes a full probability estimation problem, whose solution can be obtained via maximum a posteriori (MAP), integrated nested Laplace approximation-stochastic partial differential equation (INLA-SPDE), and related techniques. Spatiotemporal Bayesian data fusion (STBDF) casts the fusion problem as MAP estimation, defines space–time priors and observation likelihoods, and performs unlimited fusion of multiple high- and low-resolution image pairs within a Bayesian framework [19]. The Bayesian hierarchical model in [20] leverages INLA-SPDE for efficient fitting and unifies in situ measurements with satellite observations through spatial and temporal projection matrices. Despite their rigorous uncertainty modeling, Bayesian fusion methods are computationally intensive, struggle with large-scale high-resolution datasets, and often underperform in regions with pronounced land-cover changes.

Learning-based spatiotemporal fusion methods can be further divided into sparse-representation-based and deep-learning-based approaches. Sparse-representation-based models first learn a pair of dictionaries for coarse- and fine-resolution image patches so that corresponding patches across resolutions can be represented by a shared set of sparse coefficients, which are then used to reconstruct high-resolution images from CR data on the prediction date. The sparse-representation-based spatiotemporal reflectance fusion model (SPSTFM) was the first sparse-representation-based fusion approach, predicting target imagery through dictionary learning, sparse-coefficient estimation, and high-resolution reconstruction [21]. The semicoupled dictionary learning spatiotemporal fusion model (SCDNTSR) adopts a framework combining nonlocal tensor sparse representation with semicoupled dictionary learning to capture nonlocal spatiotemporal–spectral structures while leveraging semicoupled sparse-coefficient mapping priors [22]. In sparse-representation-based methods, the weak coupling among dictionary learning, sparse coding, and image reconstruction leads to complex and unstable architectures. In recent years, a growing number of deep-learning-based models have provided new avenues for remote sensing spatiotemporal fusion. The spatiotemporal fusion deep convolutional neural network (STFDCNN) utilizes a two-stage deep CNN inspired by super-resolution networks to learn nonlinear mapping and spatial detail recovery between MODIS and Landsat data, applying high-pass modulation for fused imagery and integrating feature extraction, transformation, and reconstruction [23]. The deep convolutional spatiotemporal fusion network (DCSTFN) automatically learns complex spatiotemporal–spectral mapping relationships via a deep CNN and generates predicted FR images by incorporating a temporal-increment formulation that reflects surface-cover changes [24]. The enhanced deep convolutional spatiotemporal fusion network (EDCSTFN) extends DCSTFN with a multibranch hourglass autoencoder and a composite loss function to reduce blur while preserving spectral consistency [25]. To address the severe detail loss in coarse resolution imagery by traditional single-stream super-resolution methods during large-scale spatiotemporal fusion, the two-stream CNN spatiotemporal fusion model (StfNet) uses parallel spatial and temporal streams to learn features and fuses them via an adaptive-weighting strategy that ensures temporal consistency (TC) and references nearby FR textures [26]. The spatiotemporal temperature fusion network (STTFN) employs a convolutional neural architecture designed for multiscale information integration, effectively capturing the complex nonlinear relationships between CR and FR images. It further incorporates a weighting mechanism based on spatiotemporal consistency to merge the two generated predictions [27]. The spatiotemporal fusion multiscale CNN (STFMCNN) captures spatial and temporal information through a multiscale dual-stream CNN architecture and performs fusion via a local fusion module to simultaneously handle gradual and abrupt surface reflectance changes [28]. ECPW-STFN introduces discrete wavelet transform (DWT) to decompose images into low- and high-frequency sub-bands learned independently by dual-stream networks, then reconstructs high-resolution imagery using inverse DWT (IDWT) [29]. STM-STFNet proposes a dual-perspective framework that discriminatively trains spatial and temporal subnets to exploit the complementary information of high- and low-resolution data for improved detail restoration and fine-grained temporal change characterization [30]. In recent years, researchers have increasingly incorporated Transformers and generative adversarial networks (GANs) into spatiotemporal fusion tasks, spawning a suite of innovative spatiotemporal fusion models.

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For instance, STF-Trans [31] adopts an encoder–decoder Transformer architecture and leverages the self-attention mechanism to capture long-range spatiotemporal dependencies, which in turn facilitates the accurate restoration of large-scale land-cover changes in remote sensing imagery. MCBAM-GAN [32] integrates GANs with multiscale convolutions and attention mechanisms, employing adversarial training (a competitive paradigm between generator and discriminator) to generate fused images with enhanced visual quality and richer texture details. SFT-GAN [33] combines a sparse Transformer with GANs, enabling simultaneous improvements in spatial detail preservation, spectral fidelity, and computational efficiency. While Transformer-based methods excel at capturing long-range spatiotemporal correlations, they exhibit limitations in edge texture recovery and are computationally expensive. In contrast, GAN-based approaches demonstrate strengths in restoring and enhancing image texture details but often suffer from compromises in spectral consistency and TC. Furthermore, both Transformer- and GAN-based frameworks face challenges related to unstable training dynamics [34].

Hybrid spatiotemporal fusion methods flexibly combine two or more fusion approaches according to the application scenario, aiming to achieve a balanced performance in heterogeneous surfaces, abrupt change detection, and uncertainty quantification. The flexible spatiotemporal data fusion (FSDAF) integrates spectral unmixing with spatial interpolation in a hybrid framework. Compared to traditional STARFM and pure unmixing approaches, FSDAF preserves more spatial details in heterogeneous landscapes while capturing land-cover transitions [35]. The unmixing-based Bayesian spatiotemporal fusion model (ISTBDF) integrates a spectral unmixing module within a Bayesian fusion framework to leverage mixed information in coarse resolution pixels and enhance fusion accuracy [36]. Adapt lasso-fit-FC (AL-FF) employs sparse regression to build a nonlinear temporal-phase variation model, improving the ability of the weight-function-based Fit-FC fusion method to capture complex temporal dynamics [37]. However, integrating multiple modules incurs significant computational overhead, increases implementation complexity, and limits scalability to large datasets.

Although existing remote sensing spatiotemporal fusion methods have improved fused image accuracy to varying degrees, they still exhibit several limitations. First, models that extract features only at the global scale struggle to balance high-frequency edge preservation and low-frequency structural reconstruction, leading to detail loss or texture distortion. Second, existing models emphasize local convolution and often overlook global interchannel relationships, resulting in insufficient utilization of global contextual information. Third, remote sensing images frequently contain noise such as cloud and shadow artifacts, and multitemporal data include static redundancy that can interfere with fusion outcomes. Some existing models do not eliminate these factors, thereby interfering with their fusion results. In this context, we propose a novel triple-branch spatiotemporal fusion model aimed at addressing these issues. Our model incorporates a dual-stream spatial network with spatiotemporal adaptive modulation (STAM) and a temporal difference network based on depthwise separable convolutions (DSCs) to comprehensively extract spatiotemporal features from remote sensing imagery, overcoming the shortcomings of existing methods in global information extraction, texture preservation, and noise suppression. Additionally, we introduce an STAM module and a squeeze-and-excitation (SE) block to effectively suppress irrelevant noisy features while enhancing feature recognition in complex regions, ensuring semantic consistency across the image. Finally, by designing a composite loss function, we effectively mitigate object edge artifacts and enhance texture continuity, thus improving the overall quality of the fusion results. The main innovations of our model are as follows.

1) We design a triple-branch network model that leverages the spatiotemporal details incorporated into the input images to maximally obtain temporal variety information and fine spatial features, and employ an adaptive local weighted fusion strategy to produce fusion results that closely match real imagery.
2) An STAM incorporating an SE block is designed, which adaptively assesses the importance of each spatial region through multilevel partitioning and spatial weighting, and uses channel attention from the SE block for global recalibration of fused features, significantly enhancing responsiveness of the model to local textures and edges as well as the discriminative power and robustness of feature representations.
3) A temporal difference network is constructed based on DSCs, which focuses on extracting temporal change features while reducing parameter count, thereby improving computational efficiency without sacrificing accuracy.
4) A composite loss function is designed that effectively maintains high-frequency details, thereby enhancing the edge preserving capability of the fused images.

The subsequent content of this article is divided as follows. Section II describes the overall model architecture, detailed module designs, and the composite loss function; Section III presents the datasets used and experimental results; Section IV discusses the effectiveness of each module; finally, Section V summarizes the contributions of this work.

II. Methods

The model we proposed uses five images as input: the coarse resolution image C_2 at the prediction time t_2; the coarse–fine image pair C_1, F_1 at the previous time t_1; and the coarse–fine image pair C_3, F_3 at the subsequent time t_3. Based on these five input images, the model predicts the fine resolution image F_2 at time t_2. In order to study the changes in the image at different time points, we introduce the coarse difference image (DC) and the fine difference image (DF). DC helps us capture the change trend of the image at coarse resolution, providing a foundation for spatiotemporal fusion; meanwhile, DF reflects the spatiotemporal changes in the image details, helping us restore finer land-cover textures, boundaries, and small-scale changes. The DC image between two times is expressed as follows:

$$DC_{21} = C_2 - C_1 \tag{1}$$

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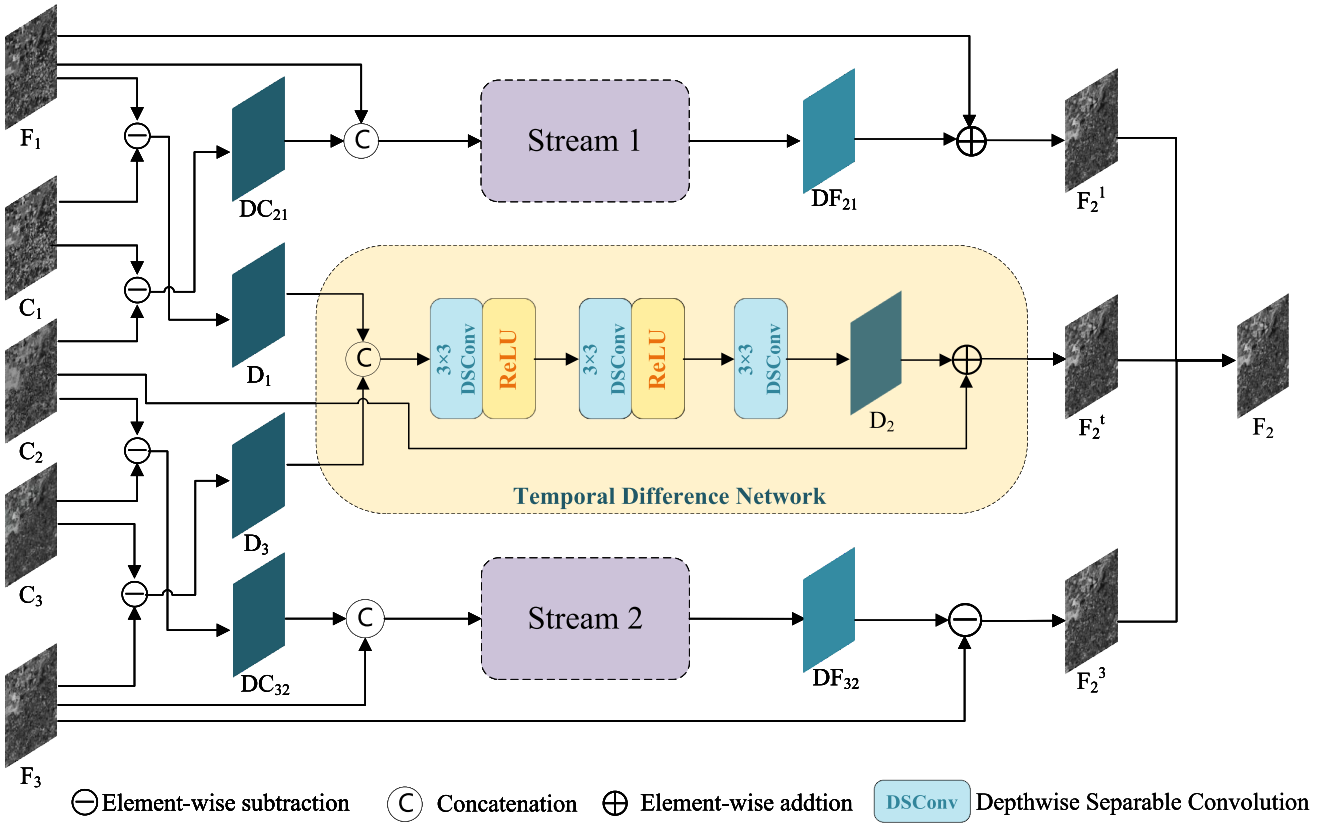


Fig. 1. Overall model framework. Define the CR and FR images at time t_i as M_i and F_i, respectively; the coarse and fine difference images between adjacent times t_i and t_j as DC_{ij} and DF_{ij} $(i > j)$, respectively; and the spatial detail information of the image at time t_i as D_i. Furthermore, F_2^1 and F_2^3 denote the fine resolution predictions at t_2 using the images at t_1 and t_3 as references, respectively, and F_2^t denotes the fine resolution prediction at t_2 generated by the temporal difference network.

$$DC_{32} = C_3 - C_2 \tag{2}$$

where DC_{21} denotes the coarse difference image between time t_2 and t_1, and DC_{32} denotes the coarse difference image between time t_3 and t_2. The DF image between two times is expressed as follows:

$$DF_{21} = F_2 - F_1 \tag{3}$$

$$DF_{32} = F_3 - F_2 \tag{4}$$

where DF_{21} denotes the fine difference image between time t_2 and t_1, and DF_{32} denotes the fine difference image between time t_3 and t_2. To improve the quality and accuracy of the final predicted image, we extracted the detail information of the fine images at each time step, which represents the spatial information of subtle changes in the image at a given time. The detail information of the fine image at each time step can be represented by the difference between the fine image and the corresponding coarse image at that time, as formulated by

$$D_1 = F_1 - C_1 \tag{5}$$

$$D_2 = F_2 - C_2 \tag{6}$$

$$D_3 = F_3 - C_3 \tag{7}$$

where D_1, D_2, and D_3 represent the detail information of the FR images at time steps t_1, t_2, and t_3, respectively.

To avoid confusion, we clearly differentiate between DC and D as follows: DC represents the difference between coarse images at different time points, reflecting the spatiotemporal change trend in the coarse images. DC helps capture the variation in the image at lower resolutions and provides the foundation for spatiotemporal fusion. Specifically, DC is used to quantify the change trends between coarse images at different time points. In contrast, D represents the difference between the fine image and the corresponding coarse image at a specific moment, capturing fine details such as edges and textures. D reflects the spatiotemporal changes at fine resolution, helping to restore detailed features and enhance fine-grained recovery.

A. Architecture

As shown in Fig. 1, the proposed model comprises three branches, including two spatial branches and one temporal branch. Specifically, the temporal branch consists of the temporal difference network, while Stream1 and Stream2 jointly form the spatial branches under the temporal dependence (TD) constraint. The two streams in the spatial branch constitute a dual-stream shared network, which aims to learn the complex mapping between coarse difference images and fine difference images of images at adjacent times. Unlike the spatial branch, the temporal branch focuses on capturing the temporal evolution of detail features of images at different times. The definition of TD is described in detail in Section II-B.

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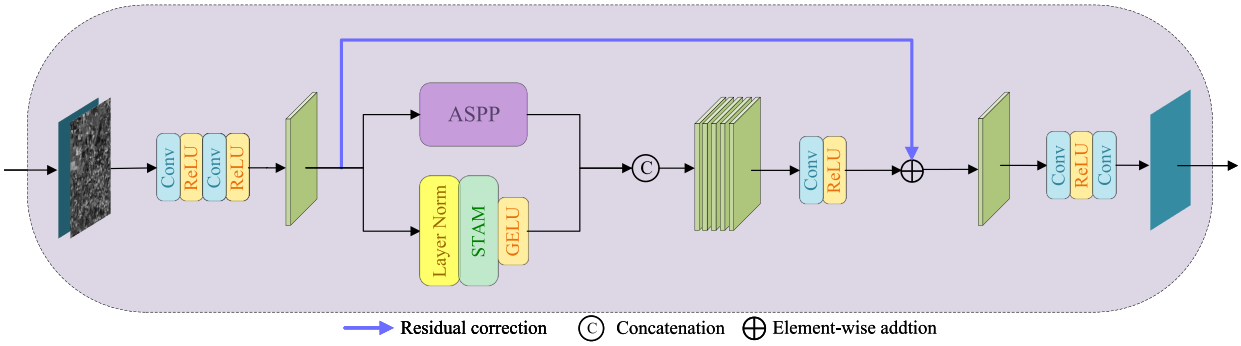


Fig. 2. Dual-stream shared network.

In the temporal branch, the detail information of the image at times t_1 and t_3 (D_1 and D_3) are concatenated and input into the temporal difference network to predict the detail information of the image D_2 at time t_2. In (6), adding D_2 to C_2 yields the unknown fine image F_2 at time t_2. The two streams in the spatial branch share the same network structure; thus, Stream1 is taken as an example. In Stream1, the coarse difference image DC_{21}, supplemented with the corresponding reference fine image F_1, is concatenated and input into the dual-stream shared network to obtain the fine difference image DF_{21} between times t_1 and t_2. According to (3), the unknown fine image F_2 at time t_2 can be obtained by adding DF_{21} to F_1. Similarly, in Stream2, according to (4), F_2 can be derived from $F_3 - DF_{32}$.

To clearly distinguish the images predicted by the three branches at time t_2, we define the following notations: F_2^1 and F_2^3 denote the unknown fine images at t_2 predicted based on the reference images at t_1 and t_3, respectively, and F_2^t denotes the fine image at t_2 generated by the temporal branch. Finally, the three images obtained from the three branches are fused via dynamic weighted fusion to produce the final prediction F_2.

B. Dual-Stream Shared Network

The two streams of the spatial branch are designed to learn the mapping from coarse resolution difference images to fine resolution difference images at corresponding times. Inspired by StfNet and to enhance the mapping capacity of the dual-stream network and its ability to learn spatial-structural similarity (SSIM), we introduce the concept of TD [26]. StfNet posits that the difference image representing temporal change is inherently correlated with the content of the corresponding original image, and it defines this correlation as TD. To integrate TD effectively into the network architecture, we supply spatial detail information via the known FR images (F_1 and F_3) at the input stage of the dual-stream shared network, thereby assisting the model in learning the mapping from difference image DC to DF.

Fig. 2 illustrates the dual-stream shared network, including ASPP and STAM blocks, which constitutes the main component of the spatial branch of the model. Due to the symmetry of the dual-stream structure, we describe Stream1 in detail as an example. In Stream1, the coarse difference image DC_{21} and the reference time fine image F_1 are concatenated and fed into the shared network to predict the fine difference image DF_{21} at time t_2. To capture richer feature information and enhance the ability of the model to represent input data details, the network incorporates a feature extraction convolutional layer that transforms the input into a 128-channel feature map. This high-channel feature map facilitates the extraction of more abundant semantic and spatial information, thereby supporting subsequent attention mechanisms and feature fusion. After feature extraction, the network employs a feature compression convolutional layer to integrate and compress the high-dimensional features into a 64-channel feature map, preparing them for the downstream modules. Both the convolutional layers for feature extraction and feature compression use 3×3 convolution kernels, which effectively capture local spatial features and facilitate the extraction of fine-grained local information from the raw images.

The compressed feature representation is then directed into two separate modules. The first module is atrous spatial pyramid pooling (ASPP), which comprises four parallel dilated convolution branches, each using a different dilation rate to capture multiscale contextual information [38]. Dilated convolutions expand the receptive field via spaced sampling without increasing parameter count, thereby capturing both local details and global context; this design enhances the stability of the model and robustness across objects of varying scales. Each dilated convolution branch uses a 3×3 convolution kernel and outputs feature maps with 64 channels, balancing the ability of the model to capture local features with computational efficiency and feature capacity, thus enhancing overall model performance. Through extensive experiments, we found that setting the four dilation rates to 6, 12, 18, and 24 yields optimal performance.

To further enhance the feature representation capacity of the model, the compressed feature map is also input to a second module. This module consists of a layer norm (LN) layer, the STAM module, and a GELU activation function. The LN layer normalizes activations within each sample, ensuring numerical stability and accelerating convergence during training. The STAM module first splits the input features into four fixed-size blocks along the channel dimension. Three of these blocks undergo an adaptive downsampling–convolution–upsampling

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pathway to capture features at multiple scales, while the fourth block retains the original resolution for direct processing. By integrating depthwise convolution with an SE attention mechanism, the module performs local adaptive weighting on these multiscale features. This design enables the model to prioritize fine-grained spatial details and selectively enhance local features [39]. A GELU activation function follows the STAM block, providing a smoother, more nuanced nonlinearity that helps capture complex feature patterns. Detailed explanations of the STAM and SE modules can be found in Sections II-B1 and II-B2.

Subsequently, the feature maps generated by four branches of ASPP and the feature map generated by the STAM module are concatenated along the channel dimension. A 1×1 convolution is then applied to merge the features from different scales into a unified output. The feature map with 320 channels is subjected to feature compression, reducing it to 64 channels. As network depth increases, gradient vanishing and explosion can occur; to mitigate this, we add the initial shared network feature map to the compressed features as a residual connection, which effectively alleviates gradient issues and enhances training stability and convergence speed. Furthermore, this residual correction preserves identity mappings, ensuring that the network retains original information and thereby improving optimization and sensitivity to subtle feature differences. Finally, two consecutive 3×3 convolutional layers expand the features back to the output channel count, producing the shared network's prediction of the fine difference image DF_{21} at time t_2. In the dual-stream shared network, all convolutional layers use ReLU activations except immediately after the STAM module, where GELU is applied.

1) STAM Block: In remote sensing spatiotemporal fusion tasks, image data typically possess high resolution, multispectral channels, and rich spatiotemporal information. However, networks with a single architecture often struggle to extract sufficient effective features from such voluminous, multiscale data. Moreover, because remote sensing images exhibit high spatial heterogeneity—with different regions showing marked differences in spectral, texture, and structural characteristics, discerning and preserving key information at various scales becomes a considerable challenge. To address this issue, we propose the STAM module, which effectively handles multiscale feature extraction and spatial heterogeneity.

The STAM module implements a spatial self-attention mechanism that processes input features in blocks and applies adaptive downsampling and upsampling strategies to different blocks, accommodating the large variation in region scales found in remote sensing imagery. This module not only captures local details but also integrates global contextual information. Specifically, each block within the STAM module undergoes depthwise convolution and SE attention, enabling adaptive weighting of regional features to emphasize critical areas while suppressing noise or background interference, thereby significantly enhancing the capacity of the model to interpret remote sensing images.

As shown in Fig. 3, the structure of the STAM module is illustrated. The feature map with 64 channels, after passing through the LN layer, is input into the STAM module and is divided into four equal parts along the channel dimension. Each part is defined as $p_i (i = 0, 1, 2, 3)$, with 16 channels, and the width and height remain consistent with the input feature map. The model employs a fixed partitioning strategy, which introduces negligible additional computational overhead while enhancing the stability of model training. To preserve local spatial information at the original scale and maintain computational efficiency for high-resolution input features, p_0 is fed directly into the SE module. Meanwhile, the other three feature groups undergo downsampling and upsampling operations with different scales before and after passing through the SE module, in order to capture information at various scales, which is then fused into a comprehensive multiscale representation. In the model implementation, the downsampling factors for p_1, p_2, and p_3 are $2\times$, $4\times$, and $8\times$, respectively. For a given input feature P, this process can be formulated as follows:

$$P \xrightarrow{\text{Split}} \left[p_0, p_1, p_2, p_3\right], \quad p_i \in \mathbb{R}^{\frac{C}{4} \times H \times W} \tag{8a}$$

$$\hat{p_0} = \text{SE}\left(p_0\right) \tag{8b}$$

$$\hat{p}_i = \text{UpSample}\left(\text{SE}\left(\text{DownSample}\left(p_i, 2^i\right)\right)\right), \quad i \in \{1, 2, 3\} \tag{8c}$$

where $\xrightarrow{\text{Split}}$ denotes the channelwise splitting operation applied to the feature map P, $\text{SE}(\cdot)$ represents the SE block, $\text{UpSample}(\cdot)$ indicates the upsampling operation that restores the feature map to its original size, $\text{DownSample}(p_i, 2^i)$ denotes the downsampling operation on the feature map with a downsampling factor of 2^i, and $\hat{p}_i (i = 0, 1, 2, 3)$ represents the output feature of each feature group.

After the above process, the features from different scales and local regions are integrated through channelwise concatenation and aggregated using a 1×1 convolution layer. This fusion strategy not only preserves key information at each scale but also provides more discriminative feature representations for subsequent spatiotemporal fusion. Finally, the features are processed by the GELU activation function and multiplied elementwise with the original input to achieve residual-style feature enhancement, further improving the representational power and the stability of the model.

2) SE Block: The structure of the squeeze-and-excitation block (SE block) is illustrated in Fig. 4. The SE block is a channel attention mechanism designed to enhance the representational capacity of the model by adaptively reweighting the features of each channel [40]. By explicitly modeling the interdependencies among channels, it dynamically adjusts the weights of feature channels to emphasize important features and better capture global contextual information. An adaptive average pooling layer is initially applied to the input feature map, compressing the spatial dimensions H and W of each channel to 1 and resulting in a global descriptor for each channel. Then, a 1×1 reduce convolution is employed to compress the channel dimensions, generating a lower dimensional representation, followed by a ReLU activation function to introduce nonlinearity into the network. Based on empirical practice, the number of channels is reduced to 1/16 of the original. Next, a 1×1 excitation convolution restores the number of channels to the original size, and a Sigmoid activation function

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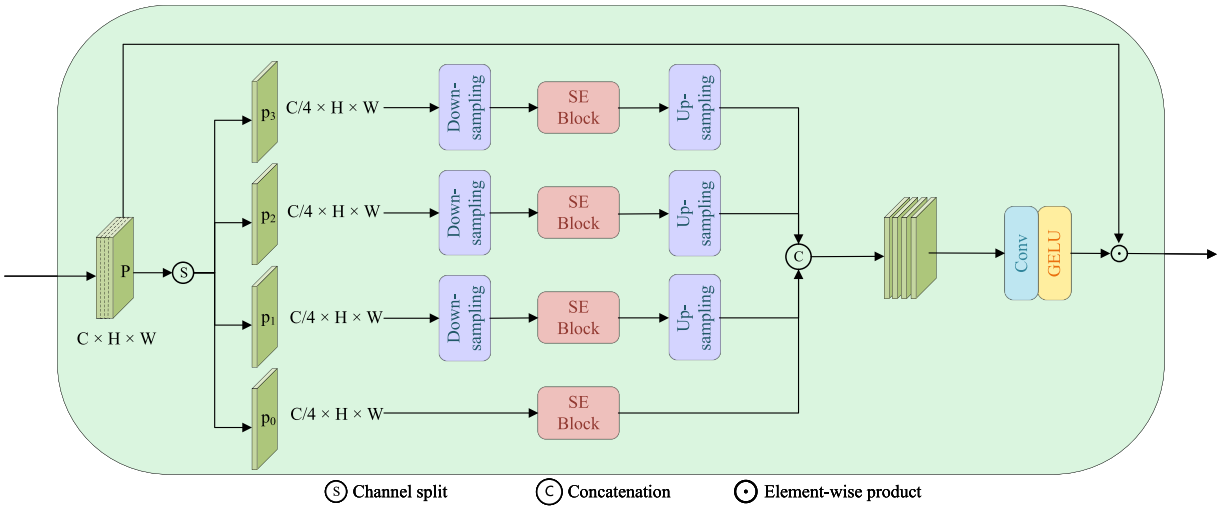


Fig. 3. STAM block.

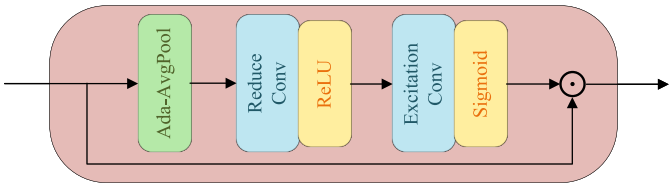


Fig. 4. SE block.

constrains the channel weights to the range [0, 1], yielding the attention weights for each channel. Finally, the input feature map is multiplied elementwise with the attention weights to produce the reweighted feature map, thereby enhancing useful features and suppressing noise.

To further enhance the sensitivity of the model to temporal changes, we introduce another key module—the temporal difference network. This module aims to refine coarse-resolution images at the target time by extracting differential features between adjacent time steps, thereby precisely capturing dynamic changes in the images. In this way, we can effectively improve the model's ability to detect fine-grained temporal variations and further optimize the performance of spatiotemporal fusion.

C. Temporal Difference Network

The core objective of this module is to efficiently and accurately refine the coarse image at the target time by extracting differential features between the coarse and fine images from preceding and succeeding time points. The differential feature extraction strategy fully exploits the temporal variation information contained in images at adjacent time steps, suppressing static structures that appear repeatedly and feeding only dynamic changes into the network. This effectively enhances the sensitivity of the model to detail changes. The module adopts a lightweight design, employing DSCs to extract temporal difference information, and finally refines the coarse image at the predicted time to generate a fine resolution result.

As shown in (5) and (7), the network computes the difference images between the fine and coarse images at times t_1 and t_3, directly extracting the changes in spatial structure and texture of remote sensing images over time. These two difference images are then concatenated along the channel dimension and input into a lightweight network composed of three DSC layers for feature extraction. The first layer applies a 3×3 depthwise convolution to independently extract spatial features within each channel, followed by a 1×1 pointwise convolution to perform channel fusion and feature mapping, expanding the feature map to 16 channels to capture rich temporal variation information. The second convolution layer maintains the 16-channel representation to further refine temporal features and enhance interchannel temporal correlations. The third layer reduces the feature dimensionality to a single channel, resulting in the predicted difference image at time t_2. Each convolution layer is succeeded by a ReLU activation to enable more effective nonlinear feature learning.

To reduce the number of parameters and computational complexity, all convolutional layers in the temporal difference network adopt DSC. By decomposing traditional convolution operations into spatial and channelwise convolutions, the model remains lightweight and more efficient than conventional convolutional neural networks, while maintaining accuracy. It contributes to a significant acceleration in both computational efficiency and inference operations of the model [41]. These three convolutional layers form a lightweight encode–nonlinear-transform–decode structure, enabling the model to effectively capture temporal variation patterns in time-series remote sensing image fusion while maintaining a favorable balance between predictive capability and computational cost. The influence of the number of DSC layers in the temporal difference network on overall model performance is discussed in detail in Section IV-D. Finally, as indicated by (6), we get

$$F_2^t = D_2 + C_2 \tag{9}$$

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where F_2^t denotes the fine-resolution image at time t_2 generated by the temporal-difference network.

D. Local Weighted Fusion Strategy

After the three branches of the model respectively generate the fine-resolution image at time t_2, the three images are merged by a fusion network to produce the final F_2 image. Previous work employed global fusion weights calculated from the absolute mean changes in the corresponding CR image of each predicted FR image, which failed to capture the local variations at each CR pixel [26]. To address this limitation, our model adopts a pixelwise local weighted fusion strategy. This strategy applies adaptive weight allocation, combining the three F_2 images using weights that are computed locally. Specifically, for each CR pixel, the weights are calculated based on a sliding window over neighboring pixels, enabling the network to focus more on regions with smaller differences during the fusion process and thereby better preserve spatial information.

First, to ensure TC of the predicted images—meaning that the predicted F_2 images from different time references should be consistent—we avoid relying solely on a single CR-FR image pair to predict the fine-resolution image at t_2. Therefore, the images F_2^1 and F_2^3 generated by the two streams in the spatial branch are fused, and during training, their difference is minimized to ensure consistency of the predictions across the temporal sequence. According to the proposed local weighting method, the images $F_2^1$and F_2^3 generated by the two streams in the spatial branch are fused into

$$F_2^s(j) = \omega_1(i) \cdot F_2^1(j) + [1 - \omega_1(i)] \cdot F_2^3(j) \tag{10}$$

where $F_2^s(j)$ represents the value at pixel j in the fused image from the two spatial streams, $F_2^1(j)$ is the pixel value at j in the FR image predicted by the first stream, and $F_2^3(j)$ is the pixel value at j in the FR image predicted by the second stream. The local fusion weight $\omega_1(i)$ is computed according to

$$\omega_1(i) = \frac{\frac{1}{|DC_{21}(i)|}}{\frac{1}{|DC_{21}(i)|} + \frac{1}{|DC_{32}(i)|}} \tag{11}$$

where $\omega_1(i)$ denotes the local fusion weight of the CR pixel i corresponding to the FR pixel j in the image F_2^1. $DC_{21}(i)$ represents the average value of all pixels within a sliding window centered at pixel i in the coarse difference image DC_{21}, and $DC_{32}(i)$ is similarly defined for DC_{32}. Therefore, in this local fusion method, all FR pixels falling within the same CR pixel share the same fusion weight.

Subsequently, the predicted image F_2^s from the spatial branch and the predicted image F_2^t from the temporal branch are locally fused to generate the final fine-resolution image F_2 at time t_2. The fusion process is defined in the following equation:

$$F_2(j) = \omega_2(i) \cdot F_2^s(j) + [1 - \omega_2(i)] \cdot F_2^t(j) \tag{12}$$

where $F_2(j)$ represents the value of pixel j in the final predicted image F_2. The value $F_2^s(j)$ corresponds to the output from the spatial branch, while $F_2^t(j)$ is obtained from the temporal branch. The local fusion weight $\omega_2(i)$ is calculated using the following equation:

$$\omega_2(i) = \frac{\frac{1}{|DC_{21}(i)+DC_{32}(i)|}}{\frac{1}{|DC_{21}(i)+DC_{32}(i)|} + \frac{1}{|D_1(i)+D_3(i)|}} \tag{13}$$

where $\omega_2(i)$ indicates the local fusion weight assigned to CR pixel i, which corresponds to pixel j in F_2^s. $D_1(i)$ denotes the average of all pixel values within a sliding window centered at pixel i in the spatial detail map of the image associated with time t_1, and $D_3(i)$ is defined in the same manner for the image at time t_3.

E. Training and Prediction

During tasks focused on spatiotemporal fusion in remote sensing, some models are commonly trained using loss functions such as $l2$ loss or SSIM loss. The $l2$ loss, also known as mean squared error (mse) loss, offers significant advantages in tasks with high spectral fidelity requirements. It ensures global consistency in pixel values and directly reflects pixel-level errors between the predicted image and the ground truth. However, mse tends to oversmooth images during optimization, leading to the loss of high-frequency details, especially at edges and fine structures. Furthermore, as it focuses solely on pixelwise discrepancies, it fails to capture structural information effectively. In contrast, SSIM loss emphasizes structural information in images, mitigating oversmoothing to some extent and preserving more detail and texture [42]. Nevertheless, due to its single-scale nature, SSIM struggles to capture details of multiscale objects in remote sensing images (e.g., farmlands and forests), and its omission of local gradient information limits its effectiveness in enhancing high-frequency components such as edges.

To alleviate these limitations, our model is optimized using a composite loss function, which consists of three components: mse loss, MS-SSIM loss, and edge-aware loss, as defined in the following equation:

$$\mathcal{L} = \mathcal{L}_{\text{mse}} + \kappa_1 \cdot \mathcal{L}_{\text{MS-SSIM}} + \kappa_2 \cdot \mathcal{L}_{\text{Edge}} \tag{14}$$

where κ_1 and κ_2 are scalar weights. mse loss, the most commonly used pixel reconstruction loss, is defined as follows:

$$\mathcal{L}_{\text{mse}} = \frac{1}{N} \sum_{i=1}^{N} (P_i - T_i)^2 \tag{15}$$

where P_i and T_i represent the ith pixel values of the predicted and target images, respectively, and N is the total number of pixels. To compensate for the limitations of mse loss in capturing structural information, we also introduce the multiscale SSIM (MS-SSIM) loss. Unlike the single-scale SSIM loss, MS-SSIM extends the advantages of SSIM by incorporating multiscale analysis. It evaluates contrast and SSIM across multiple resolutions and aggregates the results from all scales to provide a more comprehensive assessment of image quality. In spatiotemporal fusion tasks, MS-SSIM offers a more robust constraint on the model, mitigating structural distortions that may arise from single-scale evaluations. The MS-SSIM loss retains the important formulations of SSIM,

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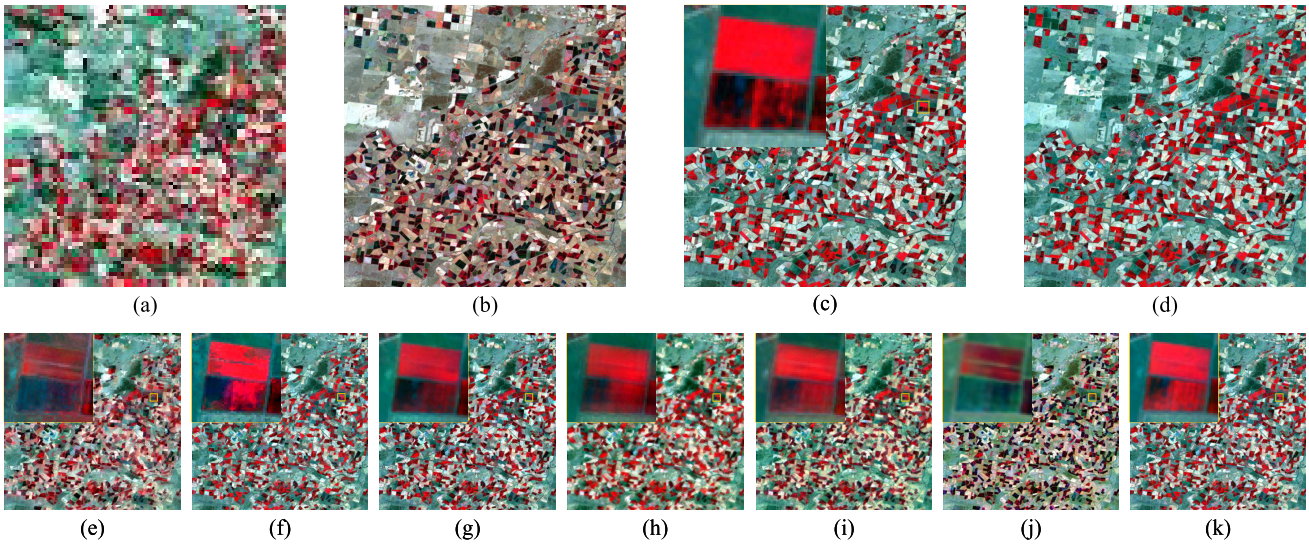


Fig. 5. Global and local prediction results for the target Landsat image in the CIA dataset (January 4, 2002). (e)–(k) Results of FSDAF, ESTARFM, STFDCNN, StfNet, STTFN, ECPW-STFN, and the proposed model, respectively. (a) MODIS image on the prediction date (January 4, 2002). (b) Landsat image on the reference date (December 3, 2001). (c) Target Landsat image on the prediction date (January 4, 2002). (d) Landsat image on another reference date (January 11, 2002).

including luminance, contrast, and structure similarity computations

$$l(P,T) = \frac{2\mu_P\mu_T + C_1}{\mu_P^2 + \mu_T^2 + C_1} \tag{16}$$

$$c(P,T) = \frac{2\sigma_{PT} + C_2}{\sigma_P^2 + \sigma_T^2 + C_2} \tag{17}$$

$$s(P,T) = \frac{\sigma_{PT} + C_3}{\sigma_P\sigma_T + C_3} \tag{18}$$

where P denotes the predicted image, and T denotes the ground truth image. μ_P and μ_T represent the mean values of images P and T, respectively. σ_P and σ_T are the standard deviations of P and T, while σ_{PT} is the covariance between them. C_1, C_2, and C_3 are stabilizing constants, where $C_1 = (0.01L)^2$, $C_2 = (0.03L)^2$, and C_3 is set to half the value of C_2. L represents the dynamic range of the image pixel values. Based on these three core components, the SSIM formula at a single scale is given as follows:

$$\mathrm{SSIM}(P,T) = \left[l(P,T)^{\alpha}\right] \cdot \left[c(P,T)^{\beta}\right] \cdot \left[s(P,T)^{\gamma}\right] \tag{19}$$

where the parameters α, β, and γ are all set to 1. MS-SSIM computes the SSIM over multiple scales, which helps better preserve the hierarchical structural information of the image

$$\mathrm{MS-SSIM}(P,T) = [l(P,T)]^{\alpha M} \cdot \prod_{j=1}^{M} [c(P,T)]^{\beta_j} \cdot [s(P,T)]^{\gamma_j} \tag{20}$$

where the weights satisfy $\alpha = \beta_j = \gamma_j$, and $j = 1, \ldots, M$. The computation of the MS-SSIM loss is defined in the following equation:

$$\mathcal{L}_{\mathrm{MS-SSIM}} = 1 - \mathrm{MS-SSIM}(P,T). \tag{21}$$

To address the limitation of mse loss in preserving high-frequency details such as edges, we incorporate edge-aware loss $\mathcal{L}_{\mathrm{Edge}}$ into the composite loss function. It measures the mse between the gradients of the predicted and target images in both horizontal and vertical directions, thereby ensuring consistency in edge and gradient transitions. This enhances high-frequency detail and effectively reduces edge blurring. The specific formulation is as follows:

$$\mathcal{L}_{\mathrm{Edge}} = \mathrm{mse}(\nabla_x P, \nabla_x T) + \mathrm{mse}\left(\nabla_y P, \nabla_y T\right) \tag{22}$$

where ∇_x and ∇_y represent the first-order gradients in the horizontal and vertical directions, respectively. These gradients are approximated by the differences between adjacent pixels along the horizontal and vertical axes, which reflect edge information and fine details in the image.

III. Experimental Results and Analysis

A. Datasets

To evaluate the performance of the proposed algorithm, we assess the robustness of the model on remote sensing datasets collected from three different time periods and geographic regions. To assess the performance of the proposed method, we perform comparative evaluations against representative models in the same domain.

The first dataset employed in this research originates from the Coleambally irrigation area (CIA) in southern New South Wales, Australia, located at 34.0034°S, 145.0675°E [43]. This region is primarily used for rice irrigation and exhibits high spatial heterogeneity, particularly in areas with relatively small irrigated farmland. The dataset contains 17 pairs of cloud-free remote sensing images acquired between October 2001 and May 2002, sourced from Landsat-7 ETM+ and MODIS Terra MOD 09 GA Collection 5. Due to the limited extent of irrigated farmland, the CIA dataset displays pronounced spatial heterogeneity, making it suitable for spatiotemporal fusion analysis.

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TABLE I
EVALUATION OF SIX OBJECTIVE INDICATORS FOR THE CIA DATASET

Evaluation	Band	Method						
		FSDAF	ESTARFM	STFDCNN	StfNet	STTFN	ECPW-STFN	Proposed
RMSE	band1	0.01525	0.01152	0.01183	0.01488	0.01313	0.01509	**0.01098**
	band2	0.02154	0.01434	0.01273	0.01414	0.01416	0.01647	**0.01096**
	band3	0.03341	0.02228	0.01835	0.02508	0.02189	0.02449	**0.01660**
	band4	0.06895	0.04075	0.03587	0.04565	0.04555	0.07338	**0.02802**
	band5	0.04671	0.04042	0.04008	0.03956	0.03685	0.04126	**0.03261**
	band6	0.03596	0.03553	0.03526	0.03851	0.03514	0.03365	**0.03051**
	Average	0.03697	0.02747	0.02569	0.02964	0.02779	0.03406	**0.02161**
SSIM	band1	0.91354	0.93468	0.94075	0.88601	0.90972	0.90605	**0.94675**
	band2	0.88126	0.92578	0.94453	0.93829	0.93793	0.92379	**0.95426**
	band3	0.80325	0.87507	0.91567	0.86459	0.89708	0.88472	**0.92565**
	band4	0.67159	0.78724	0.88273	0.82114	0.84704	0.66369	**0.89761**
	band5	0.77014	0.78153	0.83756	0.81039	0.82827	0.80814	**0.85795**
	band6	0.79895	0.79496	0.84576	0.80214	0.83397	0.83308	**0.86079**
	Average	0.80646	0.84988	0.89450	0.85376	0.87567	0.83658	**0.90717**
PSNR	band1	36.33333	38.77255	38.54144	36.54885	37.63366	36.42472	**39.19166**
	band2	33.33654	36.87140	37.90667	36.98799	36.98083	35.66749	**39.20705**
	band3	29.52209	33.04022	34.72524	32.01415	33.19367	32.22045	**35.59759**
	band4	23.22960	27.79757	28.90507	26.81115	26.83033	22.68892	**31.05189**
	band5	26.61203	27.86872	27.94118	28.05383	28.67231	27.68891	**29.73176**
	band6	28.88419	28.98911	29.05362	28.28877	29.08276	29.46141	**30.31152**
	Average	29.65296	32.22326	32.84554	31.45079	32.06559	30.69198	**34.18191**
UIQI	band1	0.86048	0.91016	0.90936	0.77579	0.79891	0.84914	**0.91827**
	band2	0.83326	**0.90524**	0.87851	0.77113	0.81556	0.86757	0.89766
	band3	0.84591	0.91385	0.92118	0.77780	0.85328	0.89162	**0.92929**
	band4	0.53043	0.85482	0.91487	0.81081	0.85752	0.80287	**0.94741**
	band5	0.88873	0.90895	0.90282	0.87244	0.90498	0.89846	**0.93010**
	band6	0.91684	0.91707	0.91802	0.87001	0.90780	0.91090	**0.93549**
	Average	0.81261	0.90168	0.90746	0.81300	0.85634	0.87009	**0.92637**
SAM	All	0.17380	0.15561	0.13777	0.12738	0.15078	0.14459	**0.11735**
ERGAS	All	2.21920	1.51153	1.51949	1.80867	2.05723	2.24423	**1.44281**

The second dataset is from the lower Gwydir catchment (LGC) in northern New South Wales, Australia (29.0855°S, 149.2815°E) [43], consisting of 14 pairs of cloud-free images from Landsat and MODIS (Terra MOD09GA Collection 5) acquired between April 2004 and April 2005. The region features extensive agricultural land and natural vegetation with relatively uniform spatial patterns. A major flood in December 2004 triggered abrupt land-cover changes, making the dataset exhibit strong temporal dynamics and particularly suitable for evaluating spatiotemporal fusion methods.

The third dataset is from the Alu Horqin Banner (AHB) in central Inner Mongolia, northeastern China, located at 43.3619°N, 119.0375°E [44]. This dataset spans from May 2013 to December 2018 and includes 27 pairs of cloud-free remote sensing images obtained from Landsat-8 OLI and MODIS Terra MOD 09 GA Collection 5. All MODIS images are from MODIS Terra MOD 09 GA Collection 5, while Landsat images are sourced from Landsat-8 OLI. Owing to its extended temporal span, the dataset captures pronounced phenological changes and diverse land-cover transitions, making the AHB dataset well-suited for spatiotemporal fusion studies.

In all three datasets, both the Landsat and MODIS images contain six key spectral bands: the blue band (0.45–0.51 μm), green band (0.53–0.59 μm), red band (0.64–0.67 μm), near-infrared band (0.85–0.88 μm), shortwave infrared 1 band (1.57–1.65 μm), and shortwave infrared 2 band (2.11–2.29 μm). For the CIA and LGC datasets, atmospheric correction of Landsat imagery was performed using the moderate resolution atmospheric transmission (MODTRAN) model, while MODIS images underwent geometric transformation [43]. In the AHB dataset, both Landsat and MODIS images were atmospherically corrected using the quick atmospheric correction (QUAC) algorithm [44]. To ensure consistency across all three datasets, Landsat images were cropped to 1200×1200 pixels, while MODIS images were cropped to 75×75 pixels and then upsampled via 16fold bilinear interpolation to match the spatial resolution of Landsat images [27].

In the experimental design, to improve training efficiency and testing accuracy, Landsat and MODIS image pairs in each band of the CIA dataset were randomly grouped into three consecutive pairs of images. During the training and validation phases, the middle image in each group was used as the reference label for loss function computation, while in the testing phase, the quality of the fused image was evaluated. To ensure the generalization capability of the model, the dataset was divided into three subsets: 60% for training, 20% for validation, and the remaining 20% for testing, with no overlap among the validation, testing, and training sets. The same partitioning strategy was applied to the LGC and AHB datasets to follow an identical training and evaluation procedure, thereby ensuring the consistency and reliability of the experimental design.

B. Evaluation

To demonstrate the effectiveness of the proposed model, we compared it with FSDAF [35], ESTARFM [17], STFDCNN

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[23], StfNet [26], STTFN [27], and ECPW-STFN [29] under the same experimental configuration, and evaluated the prediction results using six commonly used metrics.

To evaluate the fusion quality from multiple perspectives, we selected evaluation metrics from different categories, including two core dimensions: spectral fidelity and spatial SSIM, while also considering numerical accuracy and visual quality. This enables a comprehensive and objective assessment of the performance of remote sensing spatiotemporal fusion results.

The first evaluation metric is root mean square error (RMSE) [45], which is a pixel-level error statistic that reflects the absolute difference between the fused image and the reference image. Smaller deviations indicate that the predicted image is closer to the true image. The second evaluation metric is SSIM [46], which measures the spatial SSIM between two images. The SSIM value ranges from [0, 1], with values closer to 1 indicating greater similarity between the predicted and the true image. The third evaluation metric is peak signal-to-noise ratio (PSNR) [47], which is used to evaluate the overall image quality based on the ratio of maximum signal strength to noise level. A PSNR value greater than 30 dB indicates acceptable quality, while values above 40 dB are considered excellent. The fourth evaluation metric is universal image quality index (UIQI) [48], which combines correlation, luminance distortion, and contrast distortion to provide a comprehensive evaluation of image quality. Its range is [0, 1], with higher values indicating greater similarity between the two images. The fifth metric is the erreur relative globale adimensionnelle de synthése (ERGAS) [49], which considers both the spatial resolution ratio and multiband errors, and evaluates the spatial–spectral overall quality of multiband image fusion. A lower ERGAS value indicates higher image quality. The final metric is spectral angle mapper (SAM) [50], a spectral fidelity evaluation based on vector space. It calculates the angle between two spectral vectors to measure their similarity, with smaller angles indicating greater spectral similarity.

C. Analysis of Results

Under the same experimental environment, we conducted experiments on the three datasets using different spatiotemporal fusion models to evaluate the performance of the proposed model. Both objective metrics and subjective visual comparisons were used to assess the quality of the fusion results.

1) CIA Dataset: We randomly selected image data from a specific date in the CIA dataset for experimentation, and the subjective and objective evaluation results are shown in Fig. 5 and Table I, respectively. The results demonstrate that the proposed model performs well in predicting remote sensing images with pronounced spatial heterogeneity. To visually illustrate the prediction performance, Fig. 5 uses the images predicted for January 4, 2002 as an example for comparison. As shown in Fig. 5, the output from FSDAF appears globally dark, and its zoomed-in view reveals severely blurred edges and high-frequency textures, with significant spectral distortion causing a noticeable deviation in color distribution from the ground truth. Although ESTARFM achieves better global color consistency compared to FSDAF, it introduces irrelevant bright spots in shadowed areas upon magnification, and lacks sufficient spatial detail. STFDCNN preserves spatial features more effectively, but produces unnatural green artifacts in multiple regions and overly saturated edges, resulting in noticeable spectral deviation from the reference.

In StfNet, the red channel suffers from luminance bias, and the overall image appears overly smoothed, with blurred edge textures and significant loss of spatial detail. Compared to StfNet, STTFN slightly improves edge sharpness, but is still affected by excessive smoothing; green channel deviation leads to distorted vegetation rendering, and fine spatial textures are not well preserved. ECPW-STFN exhibits a generally pale tone, with visible block artifacts, significant local spectral shifts, large color errors, and blurry edges. These issues are especially evident in high-reflectance and transitional regions, making it difficult to reconstruct real-world features accurately. In contrast, the proposed model maintains a color tone that is highly consistent with the ground truth image. Only a slight contrast enhancement is applied to shadow areas to improve visual clarity. The zoomed-in view reveals well-restored spatial details and sharp edge textures, making it the closest to the original image.

Table I reports six objective metrics for the CIA dataset, with the best performance for each spectral band highlighted in bold. As shown, our method achieves the best results across all metrics, except for UIQI in band 2, where it ranks second. ESTARFM yields the highest UIQI for band 2, indicating its superior performance in capturing features in the blue wavelength. The lowest RMSE achieved by our model indicates its strong ability to accurately capture surface reflectance characteristics across different spectral ranges, thereby reducing errors caused by spatial heterogeneity. The highest SSIM confirms the superiority of the model in preserving image textures and edge-level spatial details, enabling a more realistic reconstruction of surface structures. The highest PSNR demonstrates the advantage of the model in suppressing reconstruction noise and improving image clarity. Although ESTARFM performs slightly better in handling local contrast for the blue band (band 2) in terms of UIQI, our model shows stronger overall robustness in visual fidelity. The lowest SAM indicates that our model can more accurately reconstruct spectral relationships across bands. The lowest ERGAS reflects the best overall error control across all spectral bands, further validating the generalization capability of the proposed model.

2) LGC Dataset: Fig. 6 illustrates the prediction results of each model for a specific date (July 5, 2004) in the LGC dataset. As shown, the FSDAF output suffers from blurred edges and severe loss of high-frequency details. Moreover, significant spectral shifts occur at the boundary between vegetation and bare soil, resulting in considerable deviation in color distribution from the ground truth. ESTARFM preserves some high-frequency textures but has an overall pale tone. Irrelevant bright spots appear in shadow and transition regions, indicating insufficient spectral consistency, and the zoomed-in regions show severe local distortion. STFDCNN can reconstruct the overall structural outlines relatively well but exhibits multiple purple artifacts and a generally dark tone,

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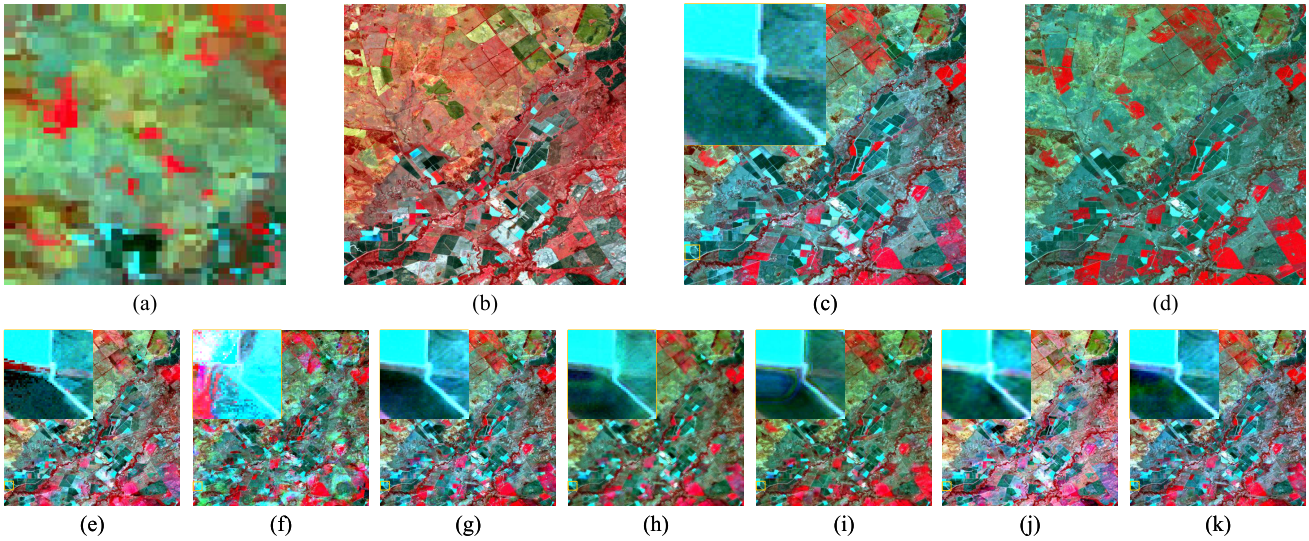


Fig. 6. Global and local prediction results for the target Landsat image in the LGC dataset (July 5, 2004). (e)–(k) FSDAF, ESTARFM, STFDCNN, StfNet, STTFN, ECPW-STFN, and the proposed model, respectively. (a) MODIS image on the prediction date (July 5, 2004). (b) Landsat image on the reference date (May 2, 2004). (c) Target Landsat image on the prediction date (July 5, 2004). (d) Landsat image on another reference date (August 6, 2004).

TABLE II
EVALUATION OF SIX OBJECTIVE INDICATORS FOR THE LGC DATASET

Evaluation	Band	Method						
		FSDAF	ESTARFM	STFDCNN	StfNet	STTFN	ECPW-STFN	Proposed
RMSE	band1	0.02322	0.03053	0.00929	0.00776	0.00795	0.00894	**0.00582**
	band2	0.01314	0.01394	0.00890	0.00985	0.00986	0.01076	**0.00730**
	band3	0.01832	0.01971	0.01425	0.01616	0.01515	0.01837	**0.01128**
	band4	0.03565	0.03528	0.02284	0.02704	0.02454	0.03247	**0.02014**
	band5	0.03079	0.03144	0.03014	0.02474	0.02282	0.02794	**0.02232**
	band6	0.03284	0.03044	0.02416	0.02334	0.02167	0.02703	**0.01899**
	Average	0.02566	0.02689	0.01826	0.01815	0.01700	0.02092	**0.01431**
SSIM	band1	0.94887	0.94465	0.97288	0.97356	0.97641	0.97386	**0.98282**
	band2	0.95358	0.94262	0.97151	0.96440	0.96811	0.96979	**0.97558**
	band3	0.91721	0.90277	0.94833	0.91931	0.94144	0.92643	**0.95254**
	band4	0.86466	0.86002	0.93525	0.90364	0.92502	0.90127	**0.93720**
	band5	0.86330	0.86810	0.90413	0.87571	0.90977	0.89554	**0.91546**
	band6	0.84602	0.86386	0.90563	0.86957	0.89856	0.88531	**0.91527**
	Average	0.89894	0.89700	0.93962	0.91770	0.93655	0.92537	**0.94648**
PSNR	band1	32.68227	30.30644	40.64131	42.20795	41.99634	40.97284	**44.69971**
	band2	37.62827	37.11562	41.01020	40.13129	40.12413	39.36028	**42.73605**
	band3	34.74297	34.10730	36.92101	35.82880	36.39458	34.71716	**38.95191**
	band4	28.95915	29.05037	32.82772	31.36007	32.20307	29.76949	**33.91967**
	band5	30.23238	30.05069	30.41578	32.13262	32.83459	31.07479	**33.02685**
	band6	29.67074	30.33172	32.33739	32.63736	33.28221	31.36337	**34.42749**
	Average	32.31930	31.82702	35.69224	35.71635	36.13915	34.54299	**37.96028**
UIQI	band1	0.86450	0.83077	0.90335	0.86924	0.90347	0.89334	**0.94179**
	band2	0.81195	0.79178	0.92472	0.86917	0.89428	0.90925	**0.93929**
	band3	0.83375	0.81474	0.92553	0.82378	0.86889	0.86901	**0.93694**
	band4	0.80650	0.82065	0.94573	0.87481	0.91178	0.86299	**0.96090**
	band5	0.89140	0.90076	0.94390	0.92159	0.94627	0.92757	**0.96176**
	band6	0.88394	0.89902	0.94814	0.92176	0.93852	0.91095	**0.96202**
	Average	0.84867	0.84295	0.93189	0.88006	0.91054	0.89552	**0.95045**
SAM	All	0.21563	0.19563	0.18150	0.19357	0.16294	0.17375	**0.15750**
ERGAS	All	1.70563	1.71842	1.08500	1.24523	1.43575	1.66290	**0.97501**

with significant distortion particularly in the blue channel. The output from StfNet is overly smoothed, with insufficient edge sharpness. Boundaries of circular and linear structures are blurred, making spatial details difficult to discern. STTFN slightly improves edge sharpness compared to StfNet but remains overly smoothed overall. The green channel shift leads to vegetation distortion, and microtextures are not effectively preserved. ECPW-STFN demonstrates relatively strong spatial information extraction, capturing more mid-frequency textures. However, the model introduces block artifacts and still shows noticeable spectral deviation along high-contrast boundaries. In comparison, the spectral information produced by our model is closest to the reference image. Shadow transitions appear natural without artifacts, edge details are clear, and the overall visual impression is superior.

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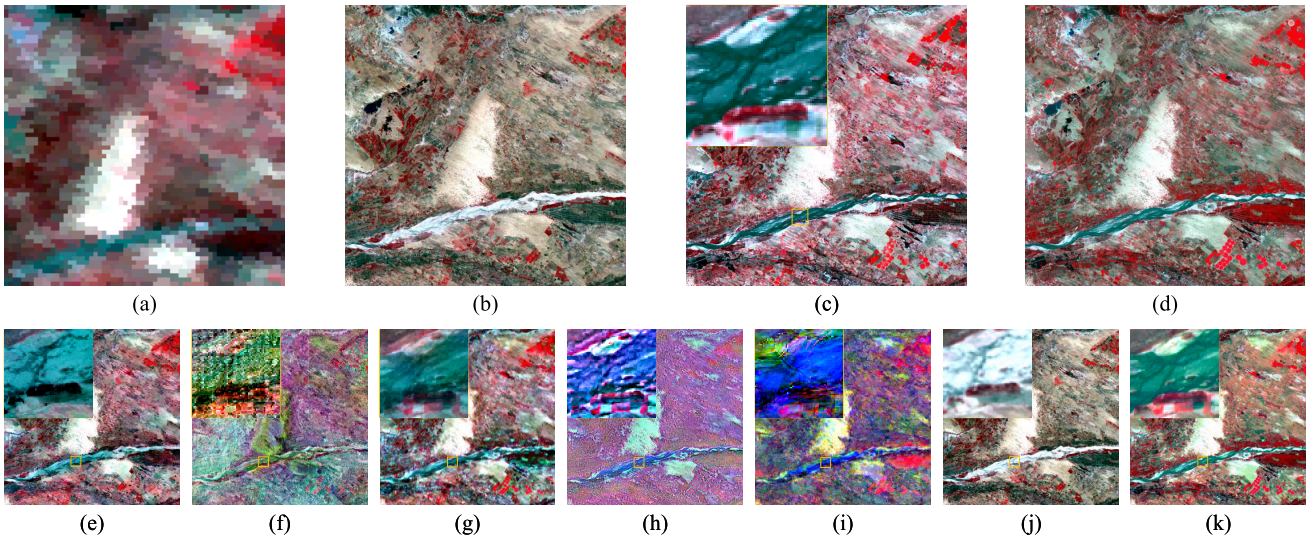


Fig. 7. Global and local prediction results for the target Landsat image in the AHB dataset (June 21, 2015). (e)–(k) FSDAF, ESTARFM, STFDCNN, StfNet, STTFN, ECPW-STFN, and the proposed model, respectively. (a) MODIS image on the prediction date (June 21, 2015). (b) Landsat image on the reference date (May 4, 2015). (c) Target Landsat image on the prediction date (June 21, 2015). (d) Landsat image on another reference date (July 7, 2015).

TABLE III
EVALUATION OF SIX OBJECTIVE INDICATORS FOR THE AHB DATASET

Evaluation	Band	Method						
		FSDAF	ESTARFM	STFDCNN	StfNet	STTFN	ECPW-STFN	Proposed
RMSE	band1	0.00256	0.00236	**0.00059**	0.00161	0.00142	0.00259	0.00069
	band2	0.00205	0.00169	0.00084	0.00168	0.00149	0.00296	**0.00062**
	band3	0.00215	0.00170	0.00075	0.00151	0.00187	0.00180	**0.00061**
	band4	0.00158	0.00349	0.00195	0.00233	0.00182	0.01054	**0.00114**
	band5	0.00247	0.00484	0.00137	0.00328	0.00250	0.00597	**0.00115**
	band6	0.00247	0.00336	0.00136	0.00262	0.00281	0.00276	**0.00118**
	Average	0.00221	0.00291	0.00114	0.00217	0.00199	0.00444	**0.00090**
SSIM	band1	0.94268	0.96489	0.99627	0.97820	0.99111	0.95620	**0.99744**
	band2	0.96591	0.98365	0.99452	0.97754	0.98597	0.94392	**0.99741**
	band3	0.96662	0.98550	0.98702	0.98458	0.97939	0.97955	**0.99810**
	band4	0.91995	0.94033	**0.99502**	0.97477	0.98553	0.87719	0.98621
	band5	0.97649	0.89134	0.99325	0.95452	0.97686	0.87675	**0.99541**
	band6	0.97229	0.94633	**0.99451**	0.96703	0.96213	0.97595	0.99312
	Average	0.95732	0.95201	0.99343	0.97277	0.98017	0.93493	**0.99462**
PSNR	band1	51.82159	52.53045	63.18579	55.86804	56.94451	51.71893	**63.22302**
	band2	53.76406	55.44733	61.48053	55.50350	56.51760	50.56025	**64.09885**
	band3	53.33814	55.41245	62.48907	56.44066	54.54969	54.88497	**64.36318**
	band4	56.03202	49.13328	**58.88021**	52.64002	54.81733	48.54062	58.86190
	band5	52.13208	46.30129	57.28147	49.68889	52.04404	44.48708	**58.78284**
	band6	52.14671	49.48092	57.34337	51.62742	51.01996	51.17319	**58.59324**
	Average	53.20577	51.38429	60.11007	53.62809	54.31552	50.22751	**61.32051**
UIQI	band1	0.72921	0.73048	0.85695	0.83476	0.82505	0.80438	**0.86048**
	band2	0.83326	0.81771	0.88296	0.84702	0.88578	0.81626	**0.91651**
	band3	0.78591	0.79985	0.81813	0.83860	0.85585	0.81777	**0.87506**
	band4	0.85043	0.83741	0.91749	0.89083	0.90452	0.86157	**0.94842**
	band5	0.80873	0.82076	0.89636	0.90209	0.84211	0.82763	**0.91123**
	band6	0.79684	0.76710	0.87488	0.84587	0.85386	0.84592	**0.89122**
	Average	0.80073	0.79555	0.87446	0.85986	0.86120	0.82892	**0.90049**
SAM	All	0.38097	0.32647	0.34449	0.22142	0.31876	0.27683	**0.19599**
ERGAS	All	2.57719	3.65906	1.87796	2.34688	3.32908	2.71286	**1.61556**

Table II presents the evaluation metrics for the LGC dataset. It can be observed that our model achieves the best performance across all spectral bands for each metric, with significantly higher average scores than all comparison methods. This indicates the strong prediction capability of the model on datasets characterized by pronounced temporal dynamics. Our model achieves the lowest RMSE values across all six bands, demonstrating more accurate reconstruction of surface reflectance. Furthermore, it attains the highest SSIM, PSNR, and UIQI values across all bands, fully reflecting its

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TABLE IV
INFLUENCE OF VARIOUS BLOCKS ON THE MODELS ON THE CIA DATASET

STAM	SE	TD Net	RMSE	SSIM	PSNR	UIQI	SAM	ERGAS
×	×	✓	0.02557	0.90107	32.73644	0.91691	0.12156	1.50861
✓	×	✓	0.02271	0.90590	33.80351	0.92136	0.11869	1.47537
✓	✓	×	0.02363	0.90323	33.39172	0.92463	0.11993	1.49663
✓	✓	✓	**0.02161**	**0.90717**	**34.18191**	**0.92637**	**0.11735**	**1.44281**

TABLE V
INFLUENCE OF VARIOUS BLOCKS ON THE MODELS ON THE LGC DATASET

STAM	SE	TD Net	RMSE	SSIM	PSNR	UIQI	SAM	ERGAS
×	×	✓	0.01839	0.94095	37.15972	0.94374	0.16542	1.13992
✓	×	✓	0.01612	0.94149	37.56771	0.94945	0.15818	0.98647
✓	✓	×	0.01758	0.94100	37.38352	0.94547	0.16100	1.11682
✓	✓	✓	**0.01431**	**0.94648**	**37.96028**	**0.95045**	**0.15750**	**0.97501**

outstanding ability to preserve spatial structural details while achieving leading performance in pixel-level accuracy, signal-to-noise ratio, and perceptual quality. The lowest SAM and ERGAS values indicate that the model reaches state-of-the-art performance in terms of spectral consistency and global error control. In summary, the proposed method outperforms all comparison models across all evaluation metrics on the LGC dataset, demonstrating excellent capability in capturing both spatial details and spectral fidelity, as well as superior generalization ability.

3) AHB Dataset: Fig. 7 presents the prediction results of each model for an image dated June 21, 2015 from the AHB dataset, which is selected as a representative case for analysis. The outputs from ESTARFM, StfNet, and STTFN show severe spectral distortions and clear deviations from the ground truth, resulting in poor overall prediction performance. FSDAF produces images with a generally dark tone and significant deviations in the red channel. It also fails to accurately capture circular spatial structures, leading to the loss of some edge details. While ECPW-STFN better preserves spectral and spatial information, its predictions rely excessively on the temporal and spatial information from the preceding reference image, resulting in a weak ability to forecast features of the target date. STFDCNN demonstrates similar color tones to the ground truth but suffers from overly blurred visual results due to excessive smoothing, leading to considerable loss of spatial details and the appearance of bright patches in multiple areas inconsistent with the original image. Although the proposed model presents slight tone differences from the reference image, it retains the most spatial detail features overall. In the zoomed-in regions, edge textures are highly restored, and the model effectively captures circular spatial patterns, achieving the best overall performance.

Table III lists the evaluation metrics for the AHB dataset. Compared with our model, STFDCNN shows relative advantages in RMSE for band 1, and in SSIM and PSNR for band 4, as well as in SSIM for band 6. These results indicate that STFDCNN is relatively sensitive to land-cover changes in these bands and performs well in reconstructing complex textures. Nevertheless, our method outperforms all others across nearly all remaining bands, demonstrating that in the AHB dataset—where land-cover changes are prominent—our model achieves the most effective predictions in scenarios requiring balanced attention to both land-cover types and high-frequency spatial details.

IV. DISCUSSION

Experimental results on the CIA dataset characterized by strong spatial heterogeneity, the LGC dataset with pronounced spatial mutations, and the AHB dataset with significant phenological variations demonstrate that the proposed model achieves superior performance and generates predictions most consistent with the ground truth. To further validate the effectiveness of each component within the model, a series of ablation experiments were conducted to assess the contribution of the key modules and to explore the impact of the composite loss function on model performance. The best results in all ablation experiments are highlighted in bold in the table, and "TDNet" refers to the temporal difference network.

A. Effectiveness of STAM Module

The experimental results indicate that removing any single module leads to a degradation in model performance, with the most noticeable decline occurring when the STAM module is excluded. As shown in Table IV, on the CIA dataset, the removal of STAM results in a significant increase in RMSE and SAM, along with a marked decrease in PSNR, indicating that this module is critical to restore the edges of complex object contours. As presented in Table V, on the LGC dataset, the introduction of STAM leads to clear improvements in metrics such as RMSE and ERGAS, demonstrating that the STAM module enhances the expression of spatial details even within large-scale and regularly textured regions. As shown in Table VI, on the AHB dataset, the effect of the STAM module is most pronounced, suggesting that STAM is particularly crucial to distinguish fine textures and structures in

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TABLE VI
INFLUENCE OF VARIOUS BLOCKS ON THE MODELS ON THE AHB DATASET

STAM	SE	TD Net	RMSE	SSIM	PSNR	UIQI	SAM	ERGAS
×	×	✓	0.00253	0.92216	55.20541	0.86177	0.26256	2.14321
✓	×	✓	0.00106	0.97348	59.62456	0.89325	0.21493	1.70658
✓	✓	×	0.00129	0.96609	58.92843	0.88947	0.22981	1.75156
✓	✓	✓	**0.00090**	**0.99462**	**61.32051**	**0.90049**	**0.19599**	**1.61556**

TABLE VII
EFFECTIVENESS OF THE NUMBER OF CONVOLUTIONAL LAYERS IN TDNET ON THREE DATASETS

Datasets	Conv Layers	RMSE	SSIM	PSNR	UIQI	SAM	ERGAS
CIA	2	0.02312	0.90380	33.68431	0.92612	0.16467	1.69020
	3	**0.02161**	**0.90717**	**34.18191**	0.92637	**0.11735**	**1.44281**
	5	0.02208	0.90612	34.04792	**0.92650**	0.12417	1.51033
LGC	2	0.01784	0.82561	34.08741	0.92833	0.16538	1.12651
	3	**0.01431**	**0.94648**	**37.96028**	**0.95045**	**0.15750**	**0.97501**
	5	0.01545	0.93214	37.53001	0.93485	0.15912	1.02965
AHB	2	0.00157	0.97958	57.79889	0.87902	0.21208	2.01741
	3	0.00090	**0.99462**	**61.32051**	**0.90049**	**0.19599**	**1.61556**
	5	**0.00072**	0.99286	60.73258	0.89960	0.19815	1.71547

hyperspectral scenarios. In summary, the STAM model plays a central role in capturing spatiotemporal correlation features and improving information interaction across spectral bands and temporal sequences. Its absence significantly weakens the ability of the model to perform high-precision predictions.

B. Effectiveness of SE Module

The data in Table IV indicate that channel importance learning through the SE module contributes positively to maintaining spectral consistency and improving reconstruction accuracy on the CIA dataset, and this effect is also observed on the AHB dataset, as shown in Table VI. However, on the LGC dataset presented in Table V, the influence of the SE module is minimal, with only slight fluctuations in all metrics, suggesting that in scenes characterized by homogeneity and high repetitiveness of the texture, the effectiveness of channel recalibration is limited.

C. Effectiveness of TDNet

As shown in Tables IV and V, the removal of the temporal difference network leads to varying degrees of degradation across multiple metrics on the CIA and LGC datasets. This effect is particularly pronounced on the LGC dataset, which exhibits strong temporal stability, indicating that the temporal difference network plays a critical role in enhancing detail in the fusion results. However, due to the large temporal span of the AHB dataset, there may be significant differences between the reference images at the preceding and succeeding time points and the target prediction time point. As a result, the contribution of the temporal difference network is relatively limited on the AHB dataset, though it still provides a slight improvement in performance metrics, as shown in Table VI.

D. Effectiveness of the Number of Convolutional Layers in TDNet

To thoroughly analyze the impact of different convolutional layer counts on performance of the model in the temporal difference network, we conducted comparative experiments under identical conditions on three datasets: CIA, LGC, and AHB. For each dataset, a random date was selected for the experiments, specifically January 4, 2002, July 5, 2004, and June 21, 2015. Six objective evaluation metrics were chosen to comprehensively assess the experimental results, as shown in Table VII. In these experiments, we set the number of convolutional layers in TDNet to 2, 3, and 5. The experimental results in Table VII indicate that with three convolutional layers, the majority of the metrics achieved the optimal results, with one suboptimal metric in each of two datasets. This suggests that convolutional networks with fewer than three layers have limited feature extraction capability and relatively poor performance, while using five layers may lead to issues such as overfitting and redundant features, causing a decline in performance. The results demonstrate that, for remote sensing spatiotemporal fusion tasks, the relationships between features do not require excessive layers to capture, and a three-layer convolutional network structure is sufficient to effectively extract information about the temporal variation of differential features in images.

E. Effectiveness of Composite Loss Function

Since the CIA dataset contains rich edge information, complex structural features, and significant pixel variations, it is the most representative and discriminative dataset to evaluate the performance of the proposed composite loss function. Therefore, when assessing the impact of different loss terms

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TABLE VIII
INFLUENCE OF DIFFERENT LOSS FUNCTIONS ON THE CIA DATASET

$\mathcal{L}_{\mathrm{MSE}}$	$\mathcal{L}_{\mathrm{MS-SSIM}}$	$\mathcal{L}_{\mathrm{Edge}}$	RMSE	SSIM	PSNR	UIQI	SAM	ERGAS
✓	×	×	0.02199	0.90459	34.16082	0.91930	0.11847	1.45932
✓	×	✓	0.02185	0.90623	34.16724	0.92479	0.11782	1.45264
✓	✓	×	0.02173	0.90609	34.16933	0.92516	0.11801	1.44594
✓	✓	✓	**0.02161**	**0.90717**	**34.18191**	**0.92637**	**0.11735**	**1.44281**

on the model, we select data from the CIA dataset for analysis, as shown in Table VIII.

When only $\mathcal{L}_{\mathrm{mse}}$ is used, the loss function reduces to the basic $l2$ loss commonly adopted in spatiotemporal fusion models. This setting only optimizes pixel-level accuracy but lacks structural awareness and edge guidance, resulting in overly smoothed images with insufficient structural detail. Consequently, this configuration yields the worst performance across all metrics in the table, particularly those sensitive to edges and structure. Upon adding the edge-aware loss $\mathcal{L}_{\mathrm{Edge}}$, all evaluation metrics show improvement, making the model pay more attention to edge structures, thus enhancing local detail and local semantic consistency. When retaining only $\mathcal{L}_{\mathrm{mse}}$ and $\mathcal{L}_{\mathrm{MS-SSIM}}$, structure-aware metrics such as SSIM and UIQI improve significantly, and RMSE also decreases. However, the degree of enhancement in SSIM, PSNR, and SAM is less than that achieved by the edge-aware loss. When all three loss terms are combined, the model achieves relatively optimal results across all metrics.

V. CONCLUSION

To overcome the limitations of existing spatiotemporal fusion methods in multiscale feature extraction, global context modeling, and suppression of static redundancy, this article proposes a triple-branch convolutional neural network. An STAM and an SE block are introduced in the dual-stream spatial network. The STAM module divides the feature map into multiple subspatial blocks for separate processing, enhancing the expression of important features, while the SE block dynamically reweights the channelwise features, amplifying critical channels and suppressing redundant ones, thereby enhancing the discriminative ability of the model for objects and texture details in the imagery. The temporal difference network focuses on directly computing the residual details between adjacent time steps and employs lightweight convolutions to rapidly refine temporal information, capturing dynamic changes in surface features while avoiding redundant computation. In addition, a composite loss is adopted, integrating pixel reconstruction, SSIM, and edge sensitivity into the optimization objective. It eliminates the limitations of single loss functions and achieves better robustness and visual performance in the fusion result. Experimental results on three different datasets demonstrate the superiority and robustness of the proposed model. Although the performance of our model surpasses other methods, it still faces challenges in areas such as the impact of the time intervals between input image reference moments on the model's prediction results.

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